

Online Payment Fraud Detection

Enterprise Payment Fraud Detection: Behavioral Anomaly Extraction

An Advanced Exploratory Data Analysis Case Study

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| The Business Problem

- Digital payment volumes are growing, but so are attacks from coordinated thieves.
- When thieves strike, the bank suffers severe financial losses and permanently damaged customer trust.
- The biggest challenge is finding the rare "needle in a haystack" without blocking legitimate customers.

ACTIONABLE INSIGHT

We must map exactly how and when fraudsters operate to stop them proactively.

| The 6.3 Million Row Challenge

- We started with a massive 500MB dataset of 6.3 million transactions.
- Processing this entire file locally is incredibly slow and expensive.
- We solved this by extracting a smart 5% "Balanced Sample" to speed up our work.

ACTIONABLE INSIGHT

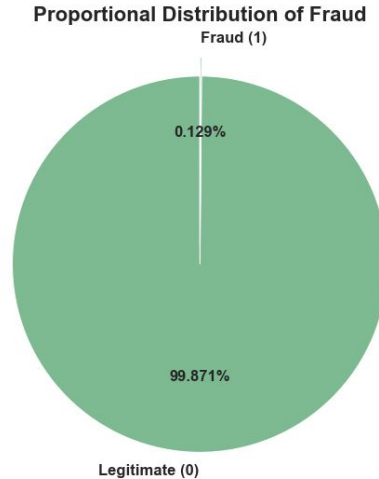
Smart sampling perfectly preserves the extreme class imbalance, allowing us to build models faster without losing any accuracy.

**6.3 Million
Transactions**



**5% Smart
Sample**

The Risk Scale



- Finding fraud is incredibly difficult because it is statistically rare.
- 99.8% of all transactions are completely safe and legitimate.
- Less than 0.2% of network traffic belongs to the thieves.

ACTIONABLE INSIGHT

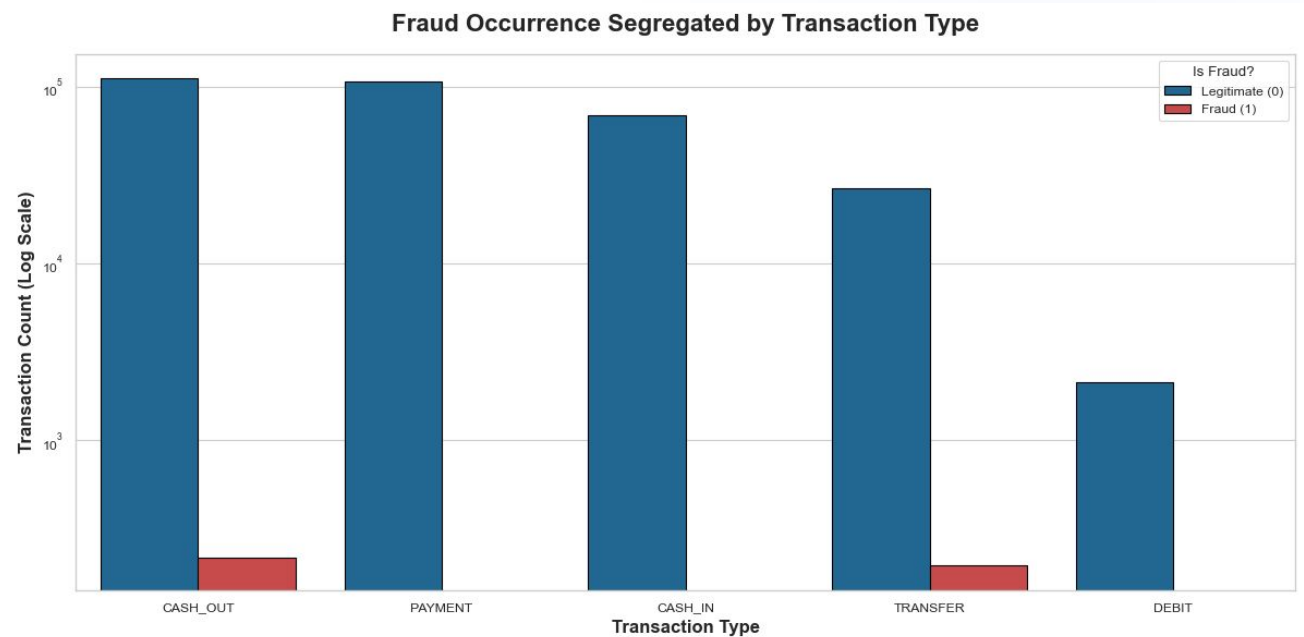
We must discard basic "Accuracy" as a metric and deploy tools specifically optimized to catch this microscopic 0.2% without bothering good customers.

The Thief's Pathways

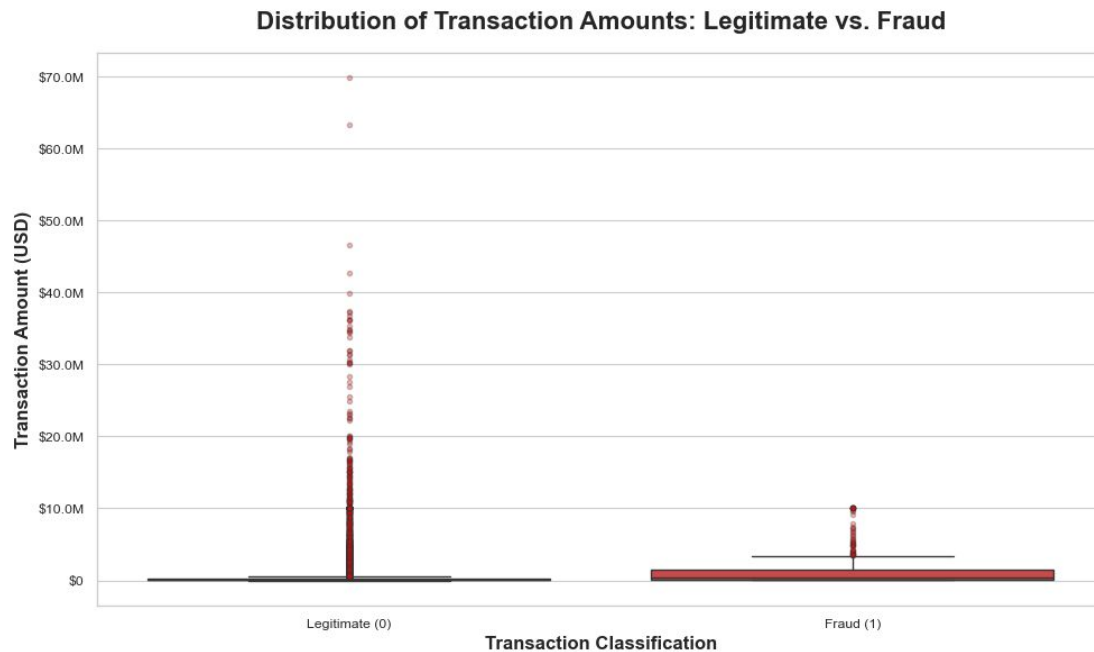
- Normal customers use all payment rails, especially regular payments and cash-ins.
- Fraudsters have a highly specific playbook, targeting only two methods.
- 100% of all fraudulent activity is isolated strictly to **'TRANSFER'** and **'CASH_OUT'**.

ACTIONABLE INSIGHT

The bank must deploy its heaviest machine learning security exclusively on TRANSFER and CASH_OUT pipelines, saving cloud resources.



The Money Gap



- Legitimate transactions mostly cluster around lower, predictable everyday amounts.
- Thieves are rarely interested in micro-transactions; they want maximum financial extraction.
- Fraudulent transfers exhibit extreme outliers, sometimes reaching multi-million dollar ranges.

ACTIONABLE INSIGHT

The Risk team must implement dynamic velocity limits, automatically routing any transfer over \$300,000 to a manual review queue.

| A New Clue

- Raw account balances alone rarely expose a thief.
- We needed to mathematically capture the operational behavior of bad actors.
- We engineered a brand-new metric: the "Account Drain Percentage".

ACTIONABLE INSIGHT

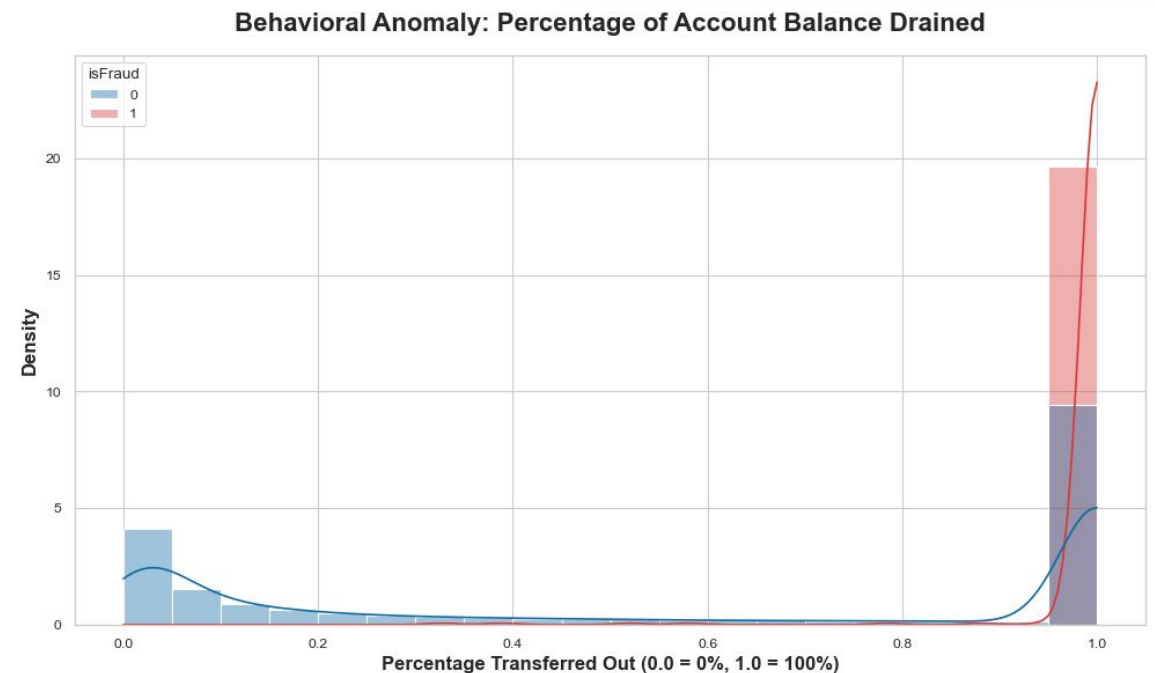
By creating derived metrics, we can capture how fast a user moves money, revealing fraud that raw data hides.

The Smoking Gun

- Normal users almost always leave some money behind after a transfer.
- Thieves maximize their illicit yield by moving funds rapidly and leaving zero trace.
- The data proves that fraudsters systematically drain **exactly 100%** of the origin account balance.

ACTIONABLE INSIGHT

Any transaction attempting to drain >95% of a total balance in a single click must instantly trigger a Multi-Factor Authentication (MFA) challenge.

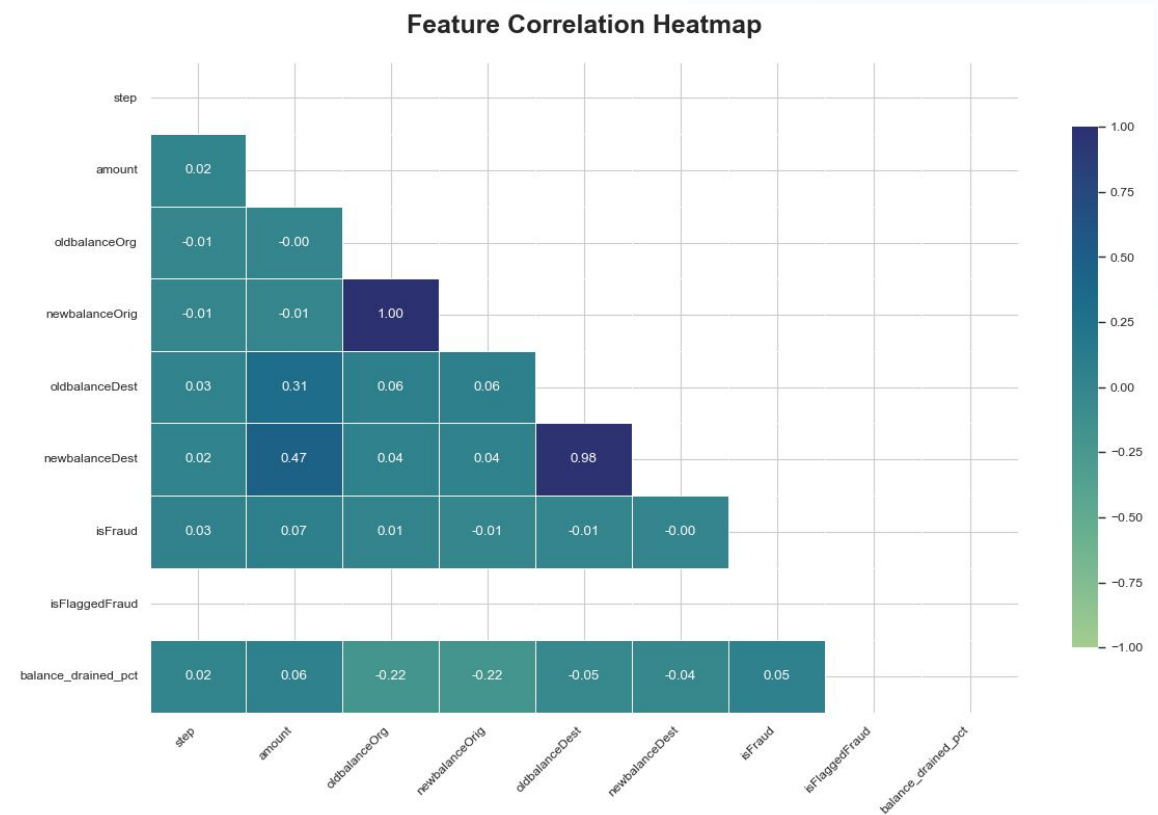


Data Connections

- No single metric can reliably predict a fraudulent event on its own.
- The separation boundaries between safe users and thieves are complex and non-linear.
- The true footprint of fraud only appears when transaction type, amount, and account drain percentage are combined.

ACTIONABLE INSIGHT

Because the boundaries are complex, basic rules will fail; we must deploy advanced Machine Learning to catch these multi-dimensional patterns.



| Next Steps to Stop Fraud

- **Deploy Targeted Friction:** Instantly challenge users attempting to drain their entire account balance.
- **Optimize Security Routing:** Focus all heavy anomaly detection models strictly on 'TRANSFER' and 'CASH_OUT' requests.
- **Upgrade to Real-Time AI:** Transition our findings into a live streaming pipeline to block thieves in sub-seconds before the money exits the ecosystem.

ACTIONABLE INSIGHT

These steps will transition the bank from reactive investigations to proactive, data-driven prevention.

Thank You

Thank you for your time.

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